M1 Where the trees put their attribution

LightGBM / XGBoost / random forest: the 36 NON-STANDARD variables hold 21.8 / 20.6 / 18.6% of the total; the HAR terms 51.7 / 52.0 / 57.6%.

Measure: share of a bucket = sum of its variables' shares; a variable's share = its mean |SHAP| (the SHAP values of all its columns summed per row, absolute value, averaged over the forecast rows) over the same summed over all 48 variables. [ ] = 95% block-bootstrap interval over origins.

$s_A = \Sigma_i |\Sigma_{j \in A} \phi_{ij}| \,/\, \Sigma_B \Sigma_i |\Sigma_{j \in B} \phi_{ij}|$ (i = forecast row, n = 33,970; j = column; A, B = the 48 variables; ϕ = TreeSHAP value under the row's origin refit)

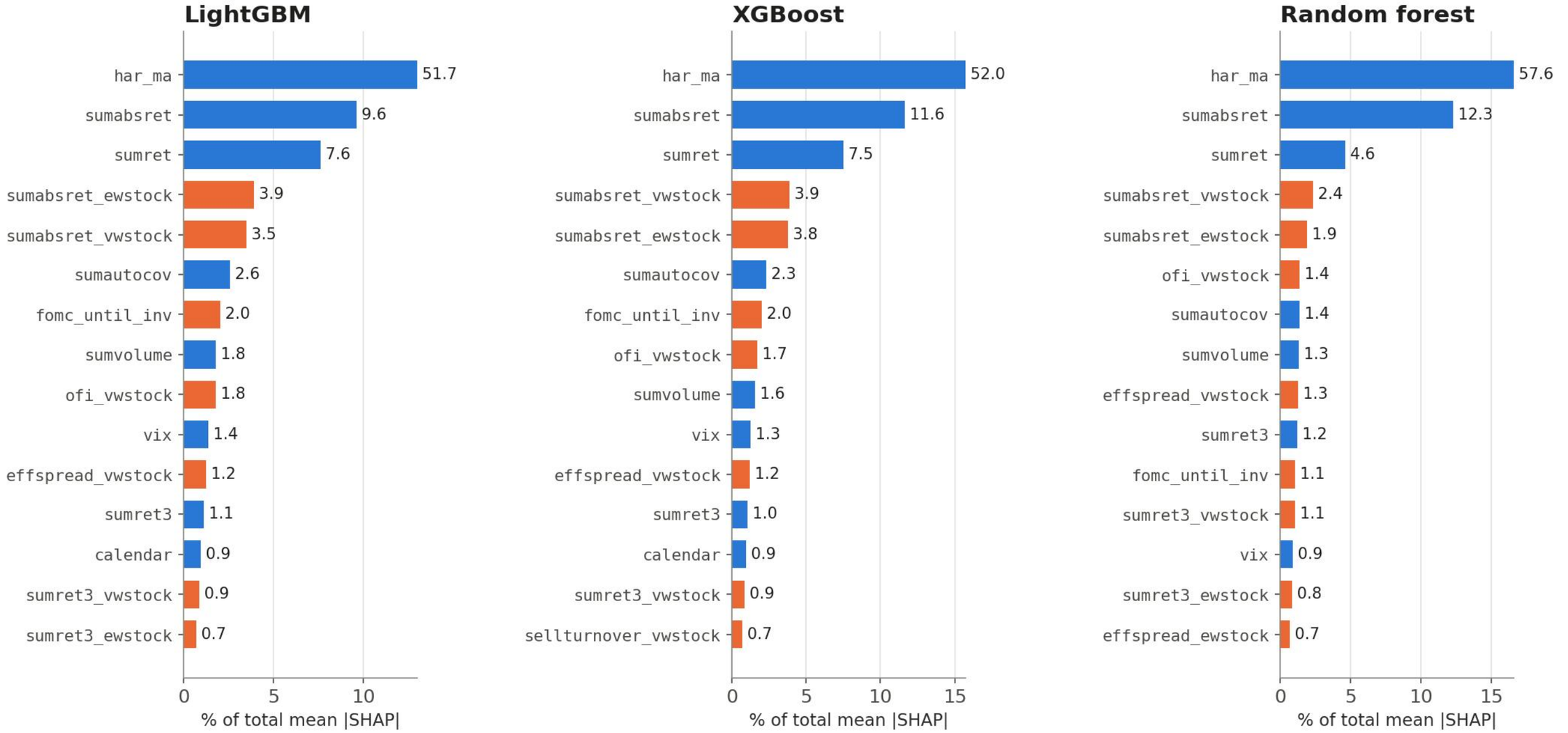
Bucket	Class	LightGBM (% of total)	XGBoost (% of total)	Random forest (% of total)
HAR terms	STANDARD	51.7 [49.5, 53.7]	52.0 [50.0, 53.8]	57.6 [55.4, 59.6]
Calendar	STANDARD	0.9 [0.8, 1.1]	0.9 [0.7, 1.1]	0.3 [0.2, 0.4]
Realized moments (E-mini)	STANDARD	22.3 [21.3, 23.4]	23.6 [22.6, 24.5]	20.7 [19.8, 21.7]
Liquidity	mixed	8.8 [8.2, 9.4]	7.9 [7.4, 8.6]	9.4 [8.8, 10.0]
Stock-market moments (EW)	NON-STD	5.5 [5.0, 6.1]	5.1 [4.6, 5.5]	3.8 [3.5, 4.2]
Stock-market moments (VW)	NON-STD	5.1 [4.5, 5.8]	5.3 [4.7, 6.0]	4.5 [4.0, 5.0]
StockTwits	NON-STD	0.8 [0.6, 0.9]	0.7 [0.6, 0.8]	0.6 [0.5, 0.7]
Implied volatility	mixed	2.3 [2.0, 2.6]	2.0 [1.7, 2.4]	1.4 [1.3, 1.6]
Volatility demand	NON-STD	0.3 [0.2, 0.4]	0.2 [0.1, 0.3]	0.2 [0.2, 0.4]
FOMC calendar	NON-STD	2.3 [2.1, 2.6]	2.2 [2.0, 2.5]	1.4 [1.2, 1.5]
STANDARD (12 variables)		**78.2 [76.6, 79.6]**	**79.4 [77.8, 80.8]**	**81.4 [79.7, 82.8]**
NON-STANDARD (36 variables)		**21.8 [20.4, 23.4]**	**20.6 [19.2, 22.2]**	**18.6 [17.2, 20.3]**

Liquidity: E-mini sumvolume / numobs (STANDARD) and the 12 constituent liquidity variables (NON-STANDARD); implied volatility: vix (STANDARD), vvix and vix3m (NON-STANDARD).

Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

M1 The variables the trees use most

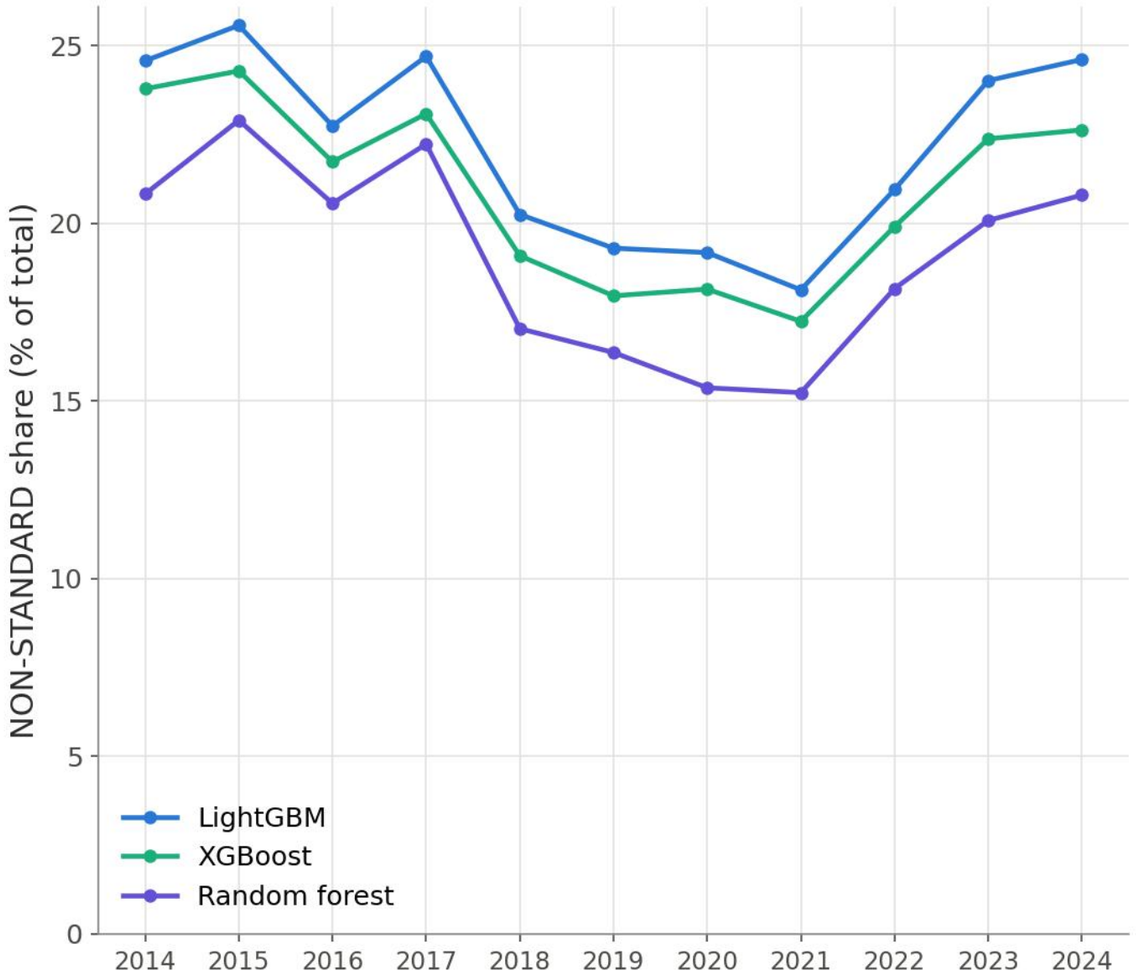
Measure: share of total mean |SHAP| per variable (all its columns summed per row), the 15 largest of each model; blue = STANDARD, orange = NON-STANDARD. Every refit's trees split on each of these variables (126 of 126). HAR-terms bar cut at the axis end.

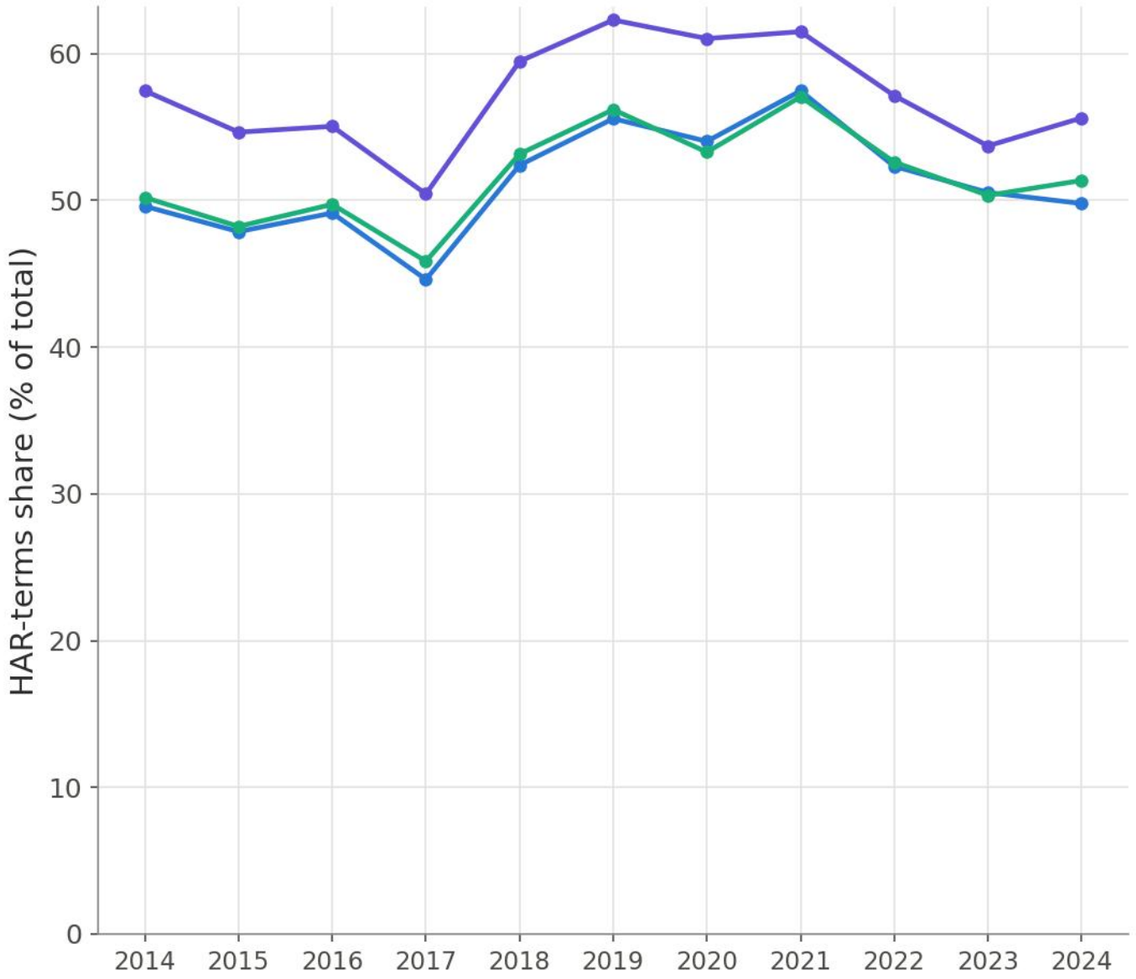


Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

M1 Non-standard share by year

Measure: the M1 shares computed on the forecast rows of each calendar year (2024: January–April). Left: NON-STANDARD share of the total; right: HAR-terms share. Each row is explained by the refit of its own origin (the origin's 21 sessions).



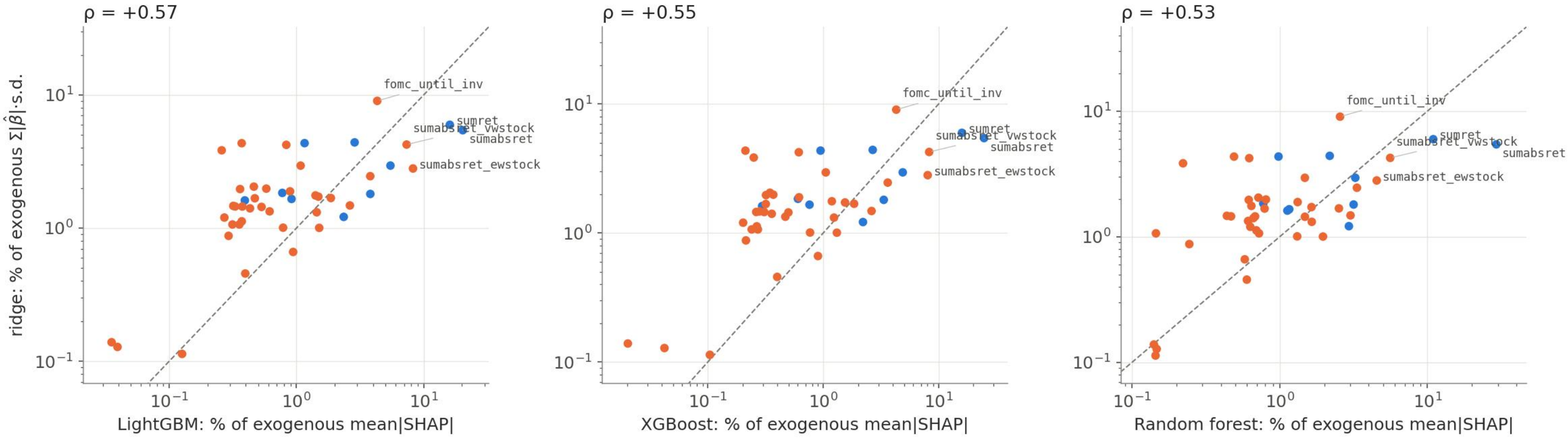


Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

M2 Trees against the linear ridge

LightGBM / XGBoost / random forest vs ridge, 46 exogenous variables: Spearman +0.57 / +0.55 / +0.53; top-10 overlap 6 / 6 / 5 of 10.

Measure: x: tree share of the exogenous total (M1 numerator over the 46 exogenous variables); y: ridge s.d.-weighted share $\Sigma_k \Sigma_{j \in A} |\hat{\beta}_{kj}| \, sd_{kj}$ / the same over the 46 exogenous variables (the W = 3,000 ridge walk at the same 126 origins). Orange = NON-STANDARD; dashed = equal shares.



NON-STANDARD share of the exogenous total, tree vs ridge (D = tree − ridge, paired block bootstrap 95%):

LightGBM: tree 45.9%, ridge 68.8%
D = −22.9 [−24.8, −21.1] pp → LESS

XGBoost: tree 43.6%, ridge 68.8%
D = −25.1 [−27.0, −23.3] pp → LESS

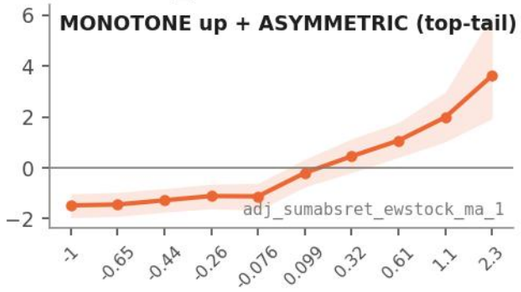
Random forest: tree 44.2%, ridge 68.8%
D = −24.5 [−26.3, −22.9] pp → LESS

Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

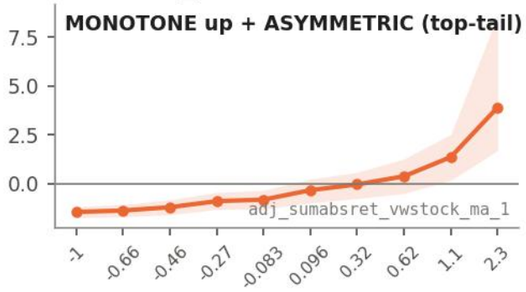
M3 Response shapes: LightGBM

Measure: mean SHAP of the variable (all columns summed) by decile of its dominant value column (the model input: trailing-window robust-scaled value), as % of the mean forecast; band = 10–90% of the row SHAP. Label = the RULE's shape class (FLAT: range < half the mean |SHAP|).

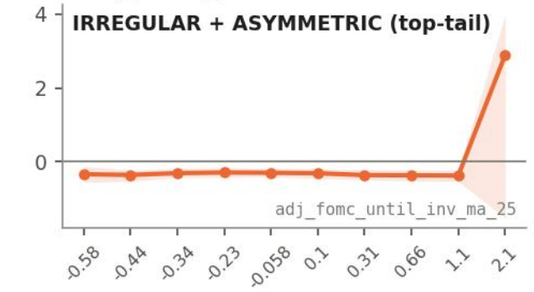
sumabsret_ewstock 3.91%



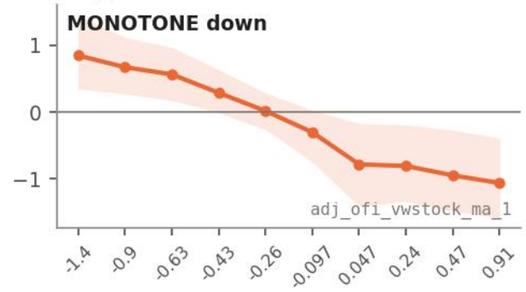
sumabsret_vwstock 3.48%



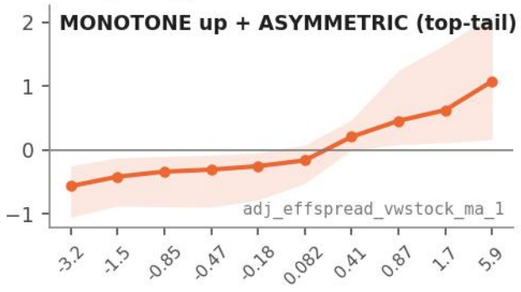
fomc_until_inv 2.05%



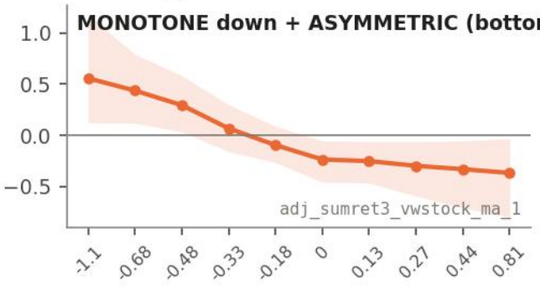
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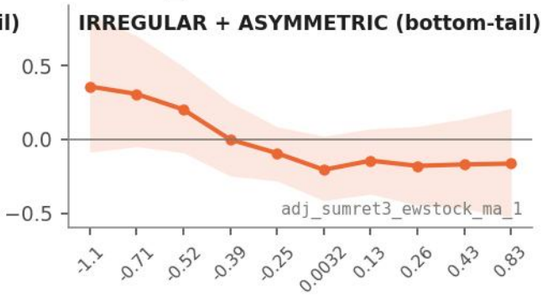
effspread_vwstock 1.25%



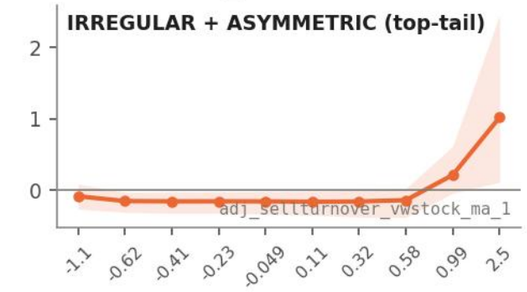
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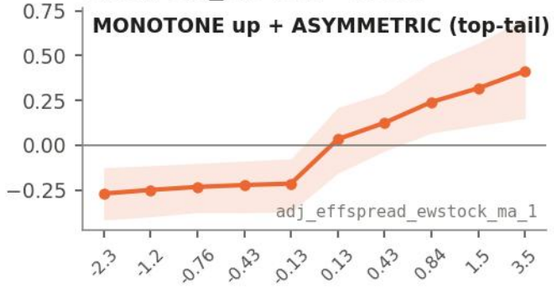
sumret3_ewstock 0.72%



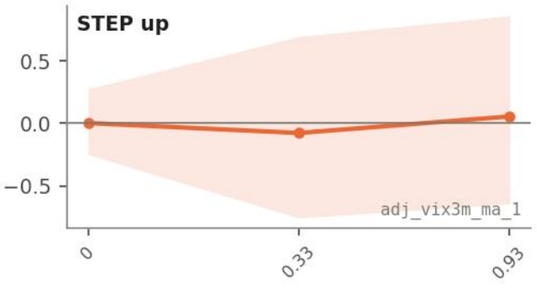
sellturnover_vwstock 0.70%



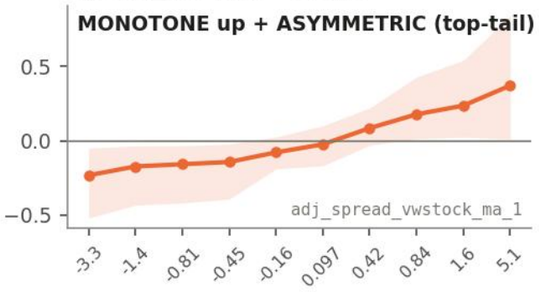
effspread_ewstock 0.68%



vix3m 0.67%



spread_vwstock 0.51%

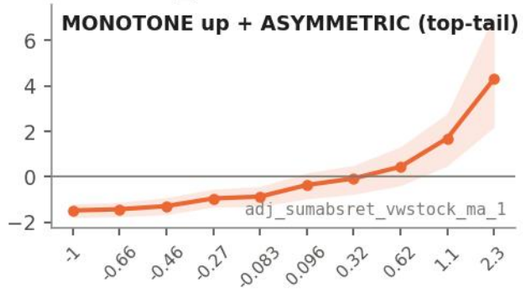


Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

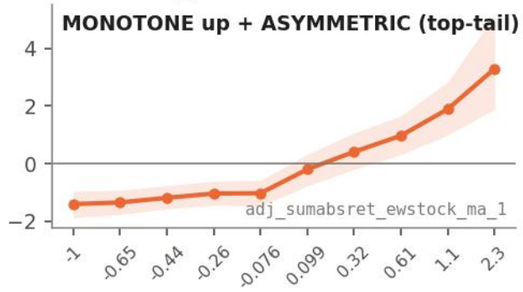
M3 Response shapes: XGBoost

Measure: mean SHAP of the variable (all columns summed) by decile of its dominant value column (the model input: trailing-window robust-scaled value), as % of the mean forecast; band = 10–90% of the row SHAP. Label = the RULE's shape class (FLAT: range < half the mean |SHAP|).

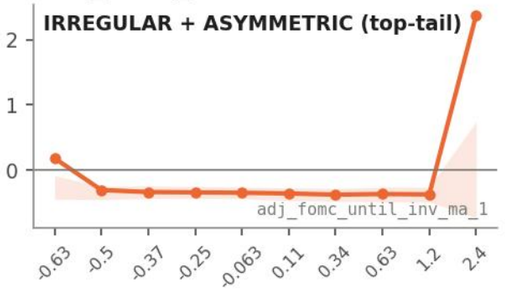
sumabsret_vwstock 3.89%



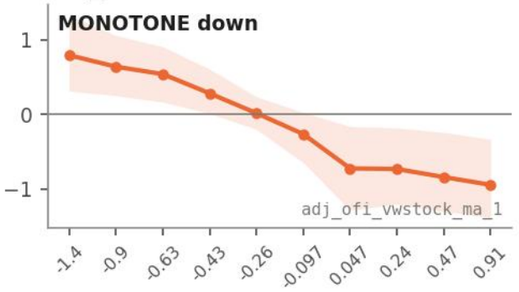
sumabsret_ewstock 3.78%



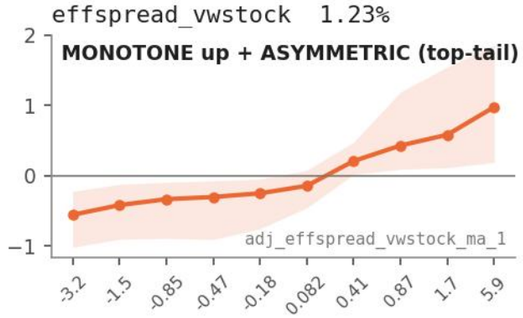
fomc_until_inv 2.02%



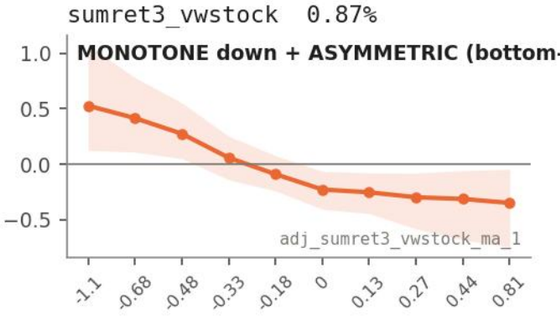
ofi_vwstock 1.70%



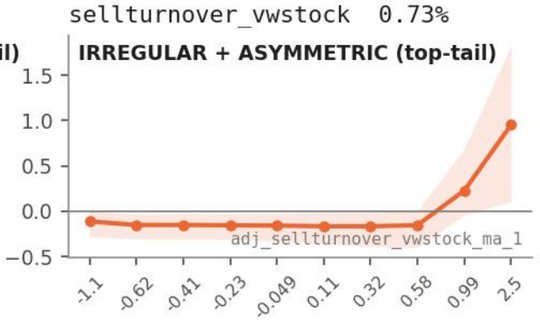
effspread_vwstock 1.23%



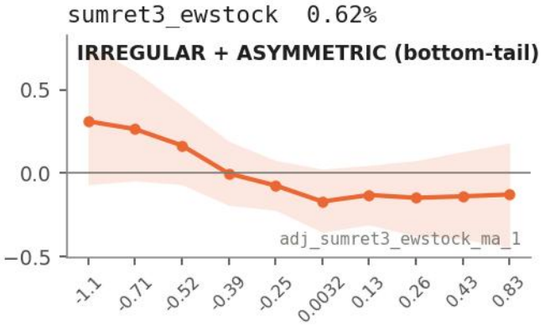
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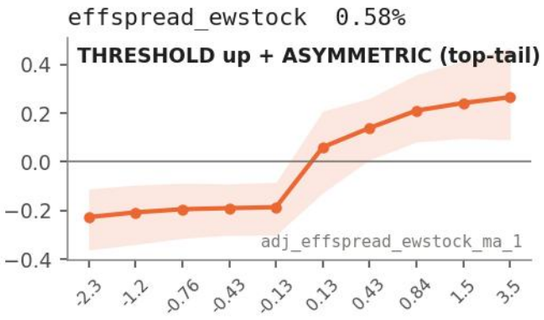
sellturnover_vwstock 0.73%



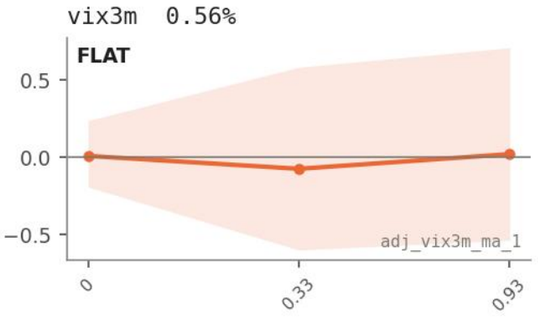
sumret3_ewstock 0.62%



effspread_ewstock 0.58%



vix3m 0.56%

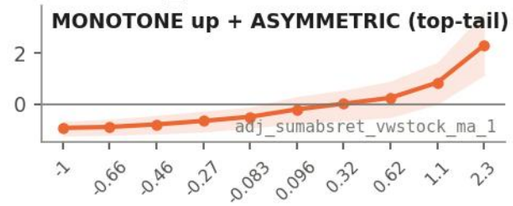


Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

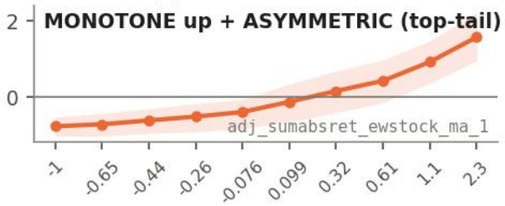
M3 Response shapes: Random forest

Measure: mean SHAP of the variable (all columns summed) by decile of its dominant value column (the model input: trailing-window robust-scaled value), as % of the mean forecast; band = 10–90% of the row SHAP. Label = the RULE's shape class (FLAT: range < half the mean |SHAP|).

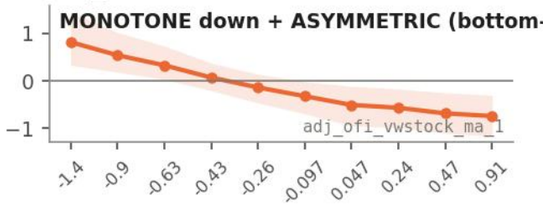
sumabsret_vwstock 2.36%



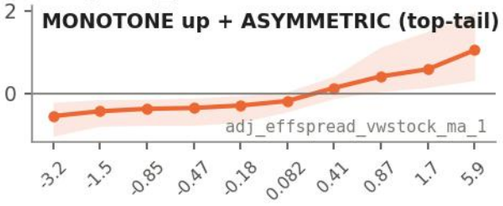
sumabsret_ewstock 1.91%



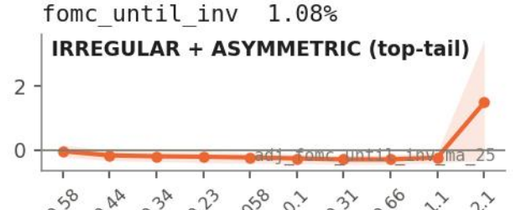
ofi_vwstock 1.40%



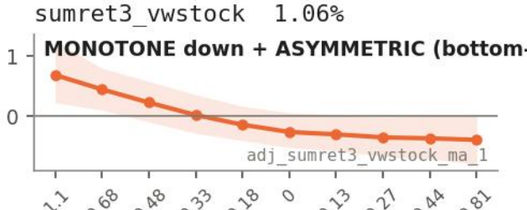
effspread_vwstock 1.27%



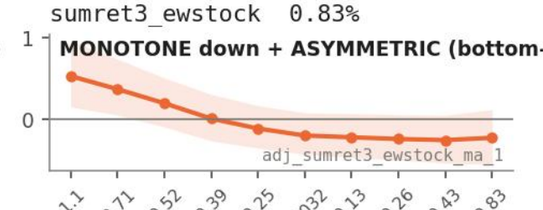
fomc_until_inv 1.08%



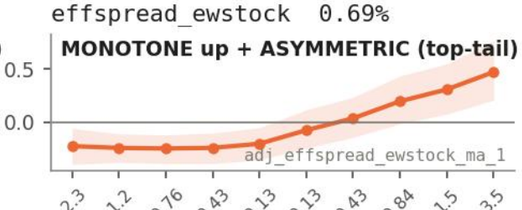
sumret3_vwstock 1.06%



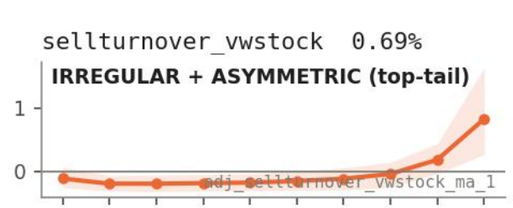
sumret3_ewstock 0.83%



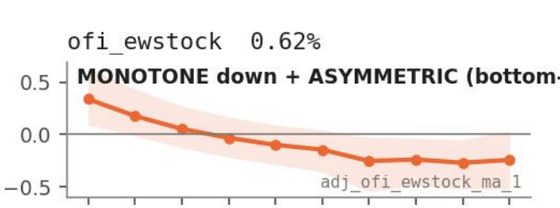
effspread_ewstock 0.69%



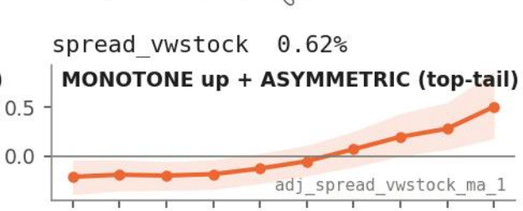
sellturnover_vwstock 0.69%



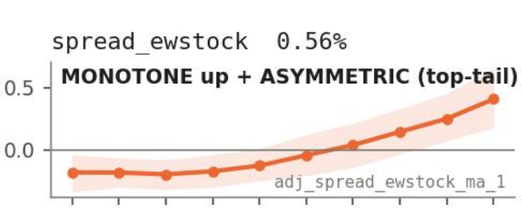
ofi_ewstock 0.62%



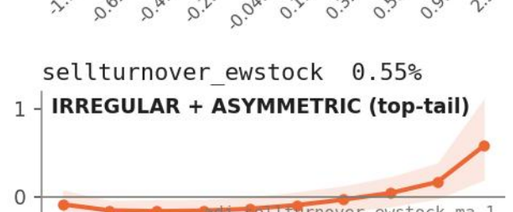
spread_vwstock 0.62%



spread_ewstock 0.56%



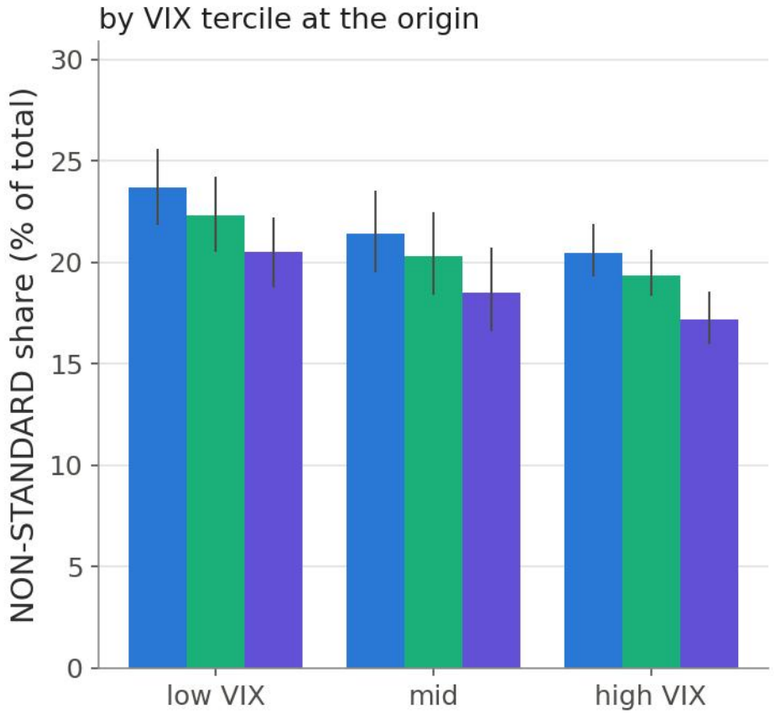
sellturnover_ewstock 0.55%

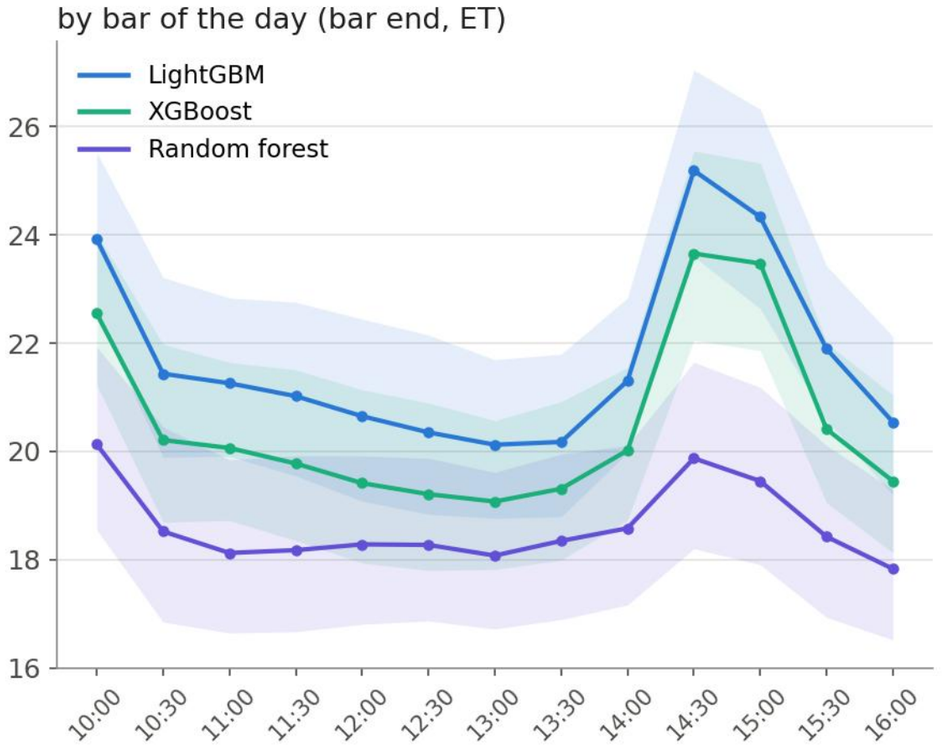


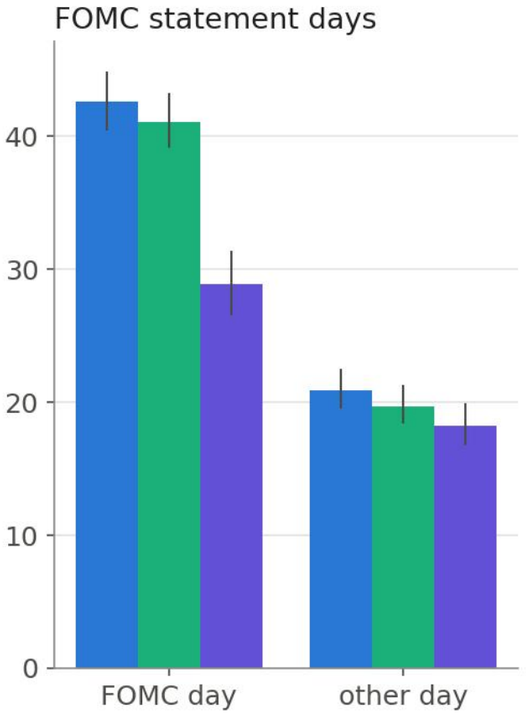
Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

M4 When do the trees use the non-standard variables?

Measure: NON-STANDARD share of total mean |SHAP| (M1) on subsets of forecast rows: by VIX tercile at the origin (prior-day close, terciles of the 126 origins), by bar of the day, and on scheduled FOMC statement days vs other days. Bars/lines: point estimate; whiskers: 95% block-bootstrap.







LightGBM: VIX high − low −3.2 pp (lower); FOMC day − other 21.7 pp (higher; FOMC-calendar bucket 26.0 pp); 16:00 − 10:00 −3.4 pp (lower)

XGBoost: VIX high − low −3.0 pp (lower); FOMC day − other 21.4 pp (higher; FOMC-calendar bucket 24.3 pp); 16:00 − 10:00 −3.1 pp (lower)

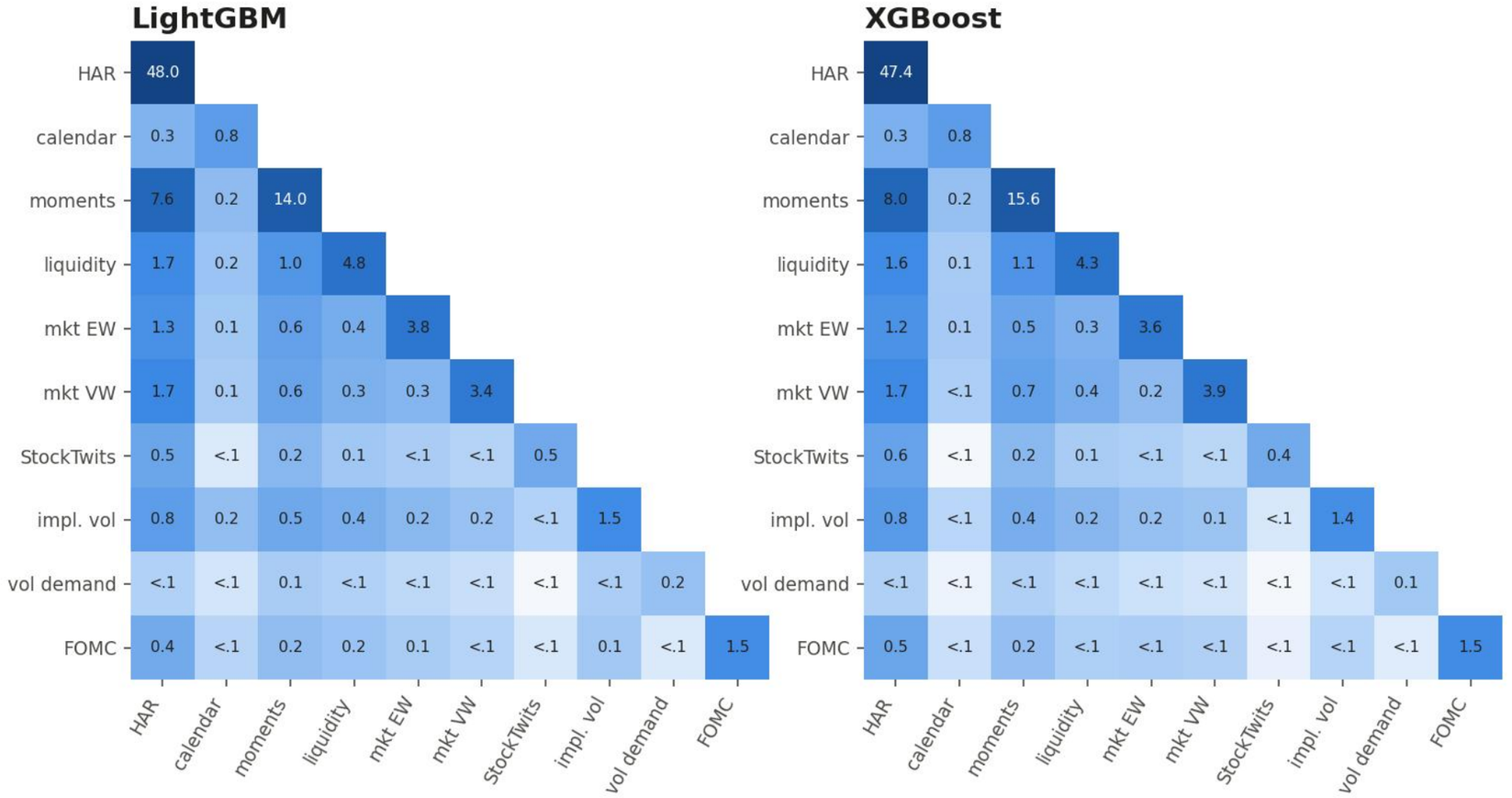
Random forest: VIX high − low −3.3 pp (lower); FOMC day − other 10.6 pp (higher; FOMC-calendar bucket 12.1 pp); 16:00 − 10:00 −2.3 pp (lower)

Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale

M5 Interactions

LightGBM / XGBoost: interactions carry 33.1 / 30.9% of all attribution (variable level);
through interactions: of a NON-STANDARD variable's attribution 49.6 / 45.7%, of a STANDARD one's 26.8 / 26.0%.

Measure: SHAP interaction values Φ_{ijl} on 1,008 fixed rows (8 a neighbourhood), summed to variables: $G_{iAB} = \Sigma_{j \in A} \Sigma_{l \in B} \Phi_{ijl}$. Cell (P, Q) of the bucket map: mean $|2\Sigma_{A \in P, B \in Q} G_{iAB}|$ (P ≠ Q) or mean $|\Sigma_{A, B \in P} G_{iAB}|$ (diagonal: main effects + within-bucket interactions), as % of the sum over the upper triangle.



Top 10 pairs with a NON-STANDARD variable (% of all pair cells)

LightGBM

```
har_ma × sumabsret_vwstock  1.25
har_ma × sumabsret_ewstock  0.86
har_ma × sellturnover_vwstock  0.31
har_ma × effspread_vwstock  0.30
har_ma × stocktwits_sentiment  0.29
har_ma × fomc_until_inv  0.26
sumret × sumabsret_ewstock  0.26
sumabsret × sumabsret_vwstock  0.25
har_ma × ofi_vwstock  0.24
sumret × sumabsret_vwstock  0.19
```

XGBoost

```
har_ma × sumabsret_vwstock  1.23
har_ma × sumabsret_ewstock  0.83
har_ma × sellturnover_vwstock  0.36
har_ma × fomc_until_inv  0.35
har_ma × effspread_vwstock  0.31
har_ma × stocktwits_sentiment  0.30
har_ma × ofi_vwstock  0.30
sumabsret × sumabsret_vwstock  0.29
sumret × sumabsret_ewstock  0.23
sumret × sumabsret_vwstock  0.22
```

Pooled trees, full set (564 columns), W = 3,000 days, 126 monthly refits 2014–2024 (the walk's refit at each origin); TreeSHAP (path-dependent) on the 33,970 forecast rows, adjusted scale